



Snowflake Advanced Tutorials — PJ's Academy

Deep, hands-on tutorials on the topics that separate senior Snowflake engineers from beginners. Each is copy-paste runnable in a Snowflake worksheet or Snowpark session.

1. Dynamic Tables — Declarative Pipelines

Dynamic Tables auto-refresh to a **target lag** — the modern replacement for many Streams+Tasks pipelines.

```
-- Raw → cleaned, refreshed automatically to within 1 minute of source
CREATE OR REPLACE DYNAMIC TABLE clean_orders
  TARGET_LAG = '1 minute'
  WAREHOUSE = etl_wh
AS
SELECT order_id, TRIM(customer) AS customer,
       ABS(amount) AS amount, order_ts
FROM raw_orders
WHERE amount IS NOT NULL;

-- Chain another dynamic table on top – it auto-maintains the DAG
CREATE OR REPLACE DYNAMIC TABLE daily_revenue
  TARGET_LAG = '5 minutes'
  WAREHOUSE = etl_wh
AS
SELECT DATE(order_ts) AS dt, SUM(amount) AS revenue
FROM clean_orders GROUP BY 1;

-- Inspect refresh history
SELECT * FROM TABLE(INFORMATION_SCHEMA.DYNAMIC_TABLE_REFRESH_HISTORY())
ORDER BY refresh_start_time DESC;
```

When to use vs Streams+Tasks: Dynamic Tables for declarative transforms (you describe the result); Streams+Tasks for imperative/complex control flow. **Know both — it's a common exam and interview question.**

2. Performance Tuning Masterclass

Diagnose with Query Profile programmatically

```
-- Find queries that spilled to disk (warehouse too small)
SELECT query_id, query_text, warehouse_size,
       bytes_spilled_to_local_storage, bytes_spilled_to_remote_storage,
       execution_time/1000 AS secs
FROM SNOWFLAKE.ACCOUNT_USAGE.QUERY_HISTORY
WHERE bytes_spilled_to_remote_storage > 0
      AND start_time > DATEADD(day,-7,CURRENT_DATE())
ORDER BY bytes_spilled_to_remote_storage DESC;
```

Clustering — measure before you cluster

```
-- Check clustering health BEFORE adding a key
SELECT SYSTEM$CLUSTERING_INFORMATION('big_events', '(event_date)');
-- Add a clustering key only if average_depth is high & table is large
ALTER TABLE big_events CLUSTER BY (event_date);
```

Search Optimization for point lookups

```
ALTER TABLE customers ADD SEARCH OPTIMIZATION ON EQUALITY(customer_id, email);
-- Great for needle-in-haystack lookups on huge tables
```

Golden rules: spilling → scale **up**; concurrency queuing → scale **out** (multi-cluster); selective filters on huge tables → **clustering** or **search optimization**; repeated aggregations → **materialized view**.

3. End-to-End RAG with Cortex (GenAI)

Build a question-answering system over your documents — entirely in Snowflake.

```
-- 1. Managed hybrid search over a knowledge base
CREATE OR REPLACE CORTEX SEARCH SERVICE kb_search
ON content
ATTRIBUTES title, url
WAREHOUSE = search_wh
TARGET_LAG = '1 hour'
AS SELECT content, title, url FROM knowledge_base;

-- 2. Retrieve + generate a grounded answer
WITH ctx AS (
  SELECT SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
    'kb_search',
    '{"query": "What is our refund policy?", "limit": 3}'
  ) AS results
)
SELECT SNOWFLAKE.CORTEX.COMPLETE(
  'llama3.1-70b',
  CONCAT('Answer using ONLY this context: ', results,
    '\nQuestion: What is our refund policy?')
) AS answer
FROM ctx;
```

This is the exact pattern behind enterprise AI assistants — retrieval grounds the LLM so it can't hallucinate.

4. Real-Time CDC Pipeline (Streams + Tasks)

```
CREATE STREAM orders_stream ON TABLE raw_orders SHOW_INITIAL_ROWS = TRUE;

CREATE OR REPLACE TASK process_orders
WAREHOUSE = etl_wh
SCHEDULE = '1 MINUTE'
WHEN SYSTEM$STREAM_HAS_DATA('orders_stream')
AS
```

```

MERGE INTO orders_summary t
USING (SELECT DATE(order_ts) dt, SUM(amount) amt
      FROM orders_stream WHERE METADATA$ACTION = 'INSERT' GROUP BY 1) s
ON t.dt = s.dt
WHEN MATCHED THEN UPDATE SET amt = t.amt + s.amt
WHEN NOT MATCHED THEN INSERT (dt, amt) VALUES (s.dt, s.amt);

ALTER TASK process_orders RESUME;  -- tasks start suspended

```

Key gotcha: tasks are created **suspended** — always `RESUME`. Use `METADATA$ACTION` / `METADATA$ISUPDATE` to distinguish inserts/updates/deletes in a stream.

5. In-Warehouse ML with Snowpark + Model Registry

```

from snowflake.ml.modeling.ensemble import RandomForestClassifier
from snowflake.ml.registry import Registry

# Train distributed inside Snowflake
model = RandomForestClassifier(
    input_cols=["tenure", "monthly_charges", "contract"],
    label_cols=["churn"], output_cols=["prediction"])
model.fit(train_df)

# Register + version the model
reg = Registry(session)
mv = reg.log_model(model, model_name="churn_model", version_name="v1",
                  metrics={"roc_auc": 0.91})

# Now score in pure SQL:
# SELECT churn_model!PREDICT(tenure, monthly_charges, contract) FROM customers;

```

Zero data egress — training, registry, and inference all live in Snowflake.

6. Cost Governance Automation

```

-- Org-wide budget guardrail
CREATE RESOURCE MONITOR org_budget WITH
  CREDIT_QUOTA = 500
  FREQUENCY = MONTHLY
  START_TIMESTAMP = IMMEDIATELY
  TRIGGERS
    ON 75 PERCENT DO NOTIFY
    ON 90 PERCENT DO NOTIFY
    ON 100 PERCENT DO SUSPEND
    ON 110 PERCENT DO SUSPEND_IMMEDIATE;

-- Chargeback report: credits per team (by warehouse naming convention)
SELECT SPLIT_PART(warehouse_name, '_', 1) AS team,
       SUM(credits_used) AS credits,
       ROUND(SUM(credits_used) * 3.0, 2) AS approx_usd  -- ~$3/credit
FROM SNOWFLAKE.ACCOUNT_USAGE.WAREHOUSE_METERING_HISTORY
WHERE start_time > DATEADD(month, -1, CURRENT_DATE())
GROUP BY 1 ORDER BY credits DESC;

```

7. Secure Data Sharing + Marketplace Enrichment

```
-- Share live data with a partner (no copy)
CREATE SHARE sales_share;
GRANT USAGE ON DATABASE analytics TO SHARE sales_share;
GRANT SELECT ON TABLE analytics.public.sales_summary TO SHARE sales_share;
ALTER SHARE sales_share ADD ACCOUNTS = partner_acct;

-- Enrich internal data with a free Marketplace dataset via a live join
SELECT s.region, s.revenue, w.avg_temp
FROM sales s
JOIN weather_marketplace.public.daily w
  ON s.region = w.region AND s.dt = w.date;
```

How to Practice

Spin up a **free Snowflake trial** (30 days, \$400 credits), create the objects above, and run each tutorial. Hands-on beats reading every time — and these are the exact scenarios the Advanced certs test.

 *Snowflake Mastery — PJ's Academy*

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